Grade 7
Unit 11
Lesson 9 Practice

Name____________________
Date____________________
Period____________________

1. Complete the table to find the area and circumference of a circle when given its diameter.

Diameter	3	9	13	18	22
Circumference					
Area					

2. Complete the table to find the area and circumference of a circle when given its radius.

Radius	2	11	23	33	36
Circumference					
Area					

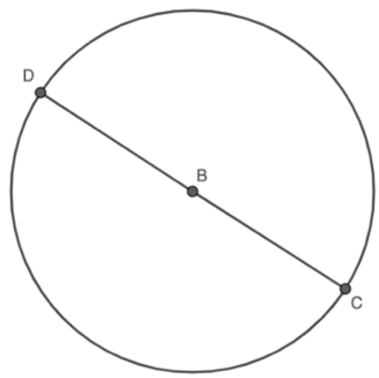
Complete the statements.

3. There
   ○ is
   ○ is not
   a proportional relationship between the diameter and circumference of a circle,
   ○ Therefore, there is no constant of proportionality.
   ○ The constant of proportionality is ______.

4. There
   ○ is
   ○ is not
   a proportional relationship between the radius and circumference of a circle.
   ○ Therefore, there is no constant of proportionality.
   ○ The constant of proportionality is ______.

5. There
   ○ is
   ○ is not
   a proportional relationship between the radius and area of a circle.
   ○ Therefore, there is no constant of proportionality.
   ○ The constant of proportionality is ______.

Circle B is shown with diameter of CD.



6. Tomas uses a drawing program to change the size of Circle B. Find the circumference, in terms of π, for each diameter he uses.

Diameter	**Circumference**
14	
25	
47	
137	

7. Suppose x represents the diameter and y represents the circumference. Determine the constant of proportionality, $\frac{y}{x}$.

8. Complete the statement.

   As the value of the diameter increases by 2, the value of circumference increases by __________.

9. Write an equation that represents the relationship between x, the diameter, and y, circumference.